

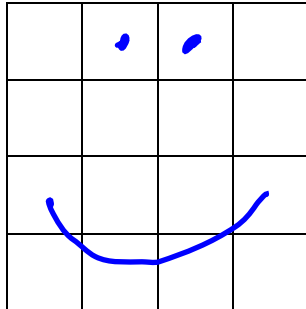


Digital Mirror

Am I friend or foe?

How does a Computer See

Sketch a **Happy** face



Write a **Happy** face as 0 and 1s

0	1	1	0
0	0	0	0
1	0	0	1
0	1	1	0

Comparing the same face using maths

Happy

0	1	1	0
0	0	0	0
1	0	0	1
0	1	1	0



Happy

0	1	1	0
0	0	0	0
1	0	0	1
0	1	1	0



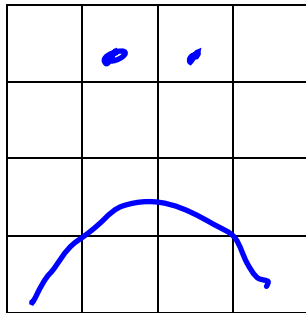
0	1	1	0
0	0	0	0
1	0	0	1
0	1	1	0

So what does Happy x Happy = **HAPPY**

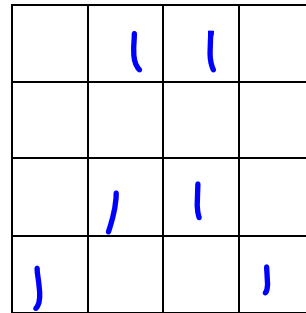
Add all the numbers together.

Happy x Happy = **6**

Sketch a sad face

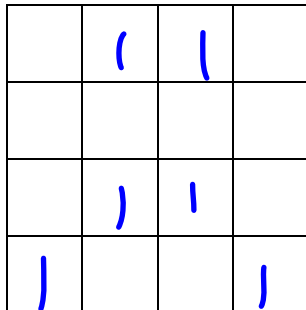


Write a sad face as numbers

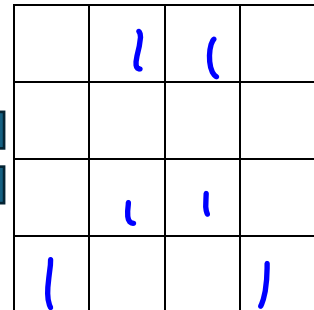
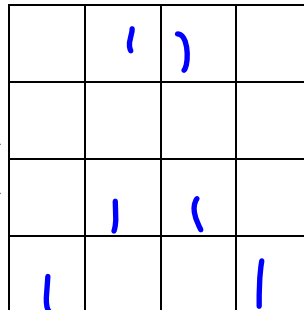


Comparing the same face using maths

Sad



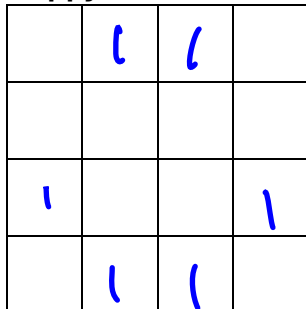
Sad



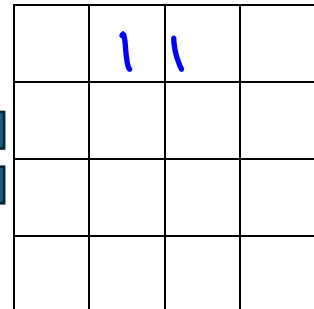
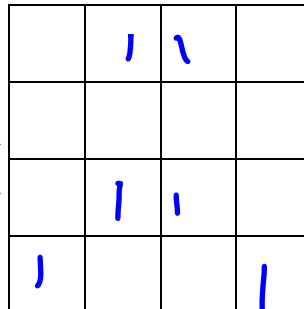
$Sad \times Sad =$ 6

Comparing the same face using maths

Happy



Sad



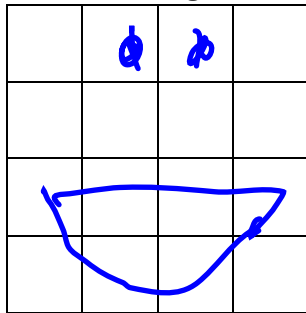
$Happy \times Sad =$ 2

What faces are most alike?

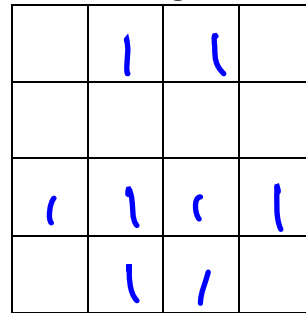
Happy, sad and Delighted



Sketch a Delighted face

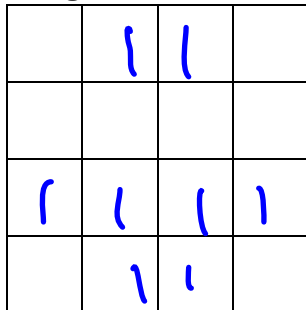


Write a Delighted face as numbers

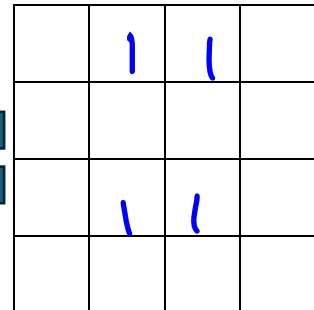
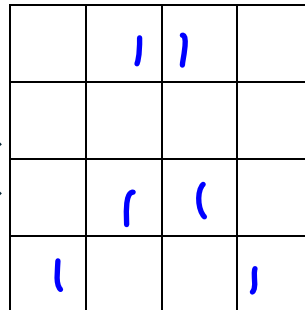


Comparing the more faces using maths

Delighted



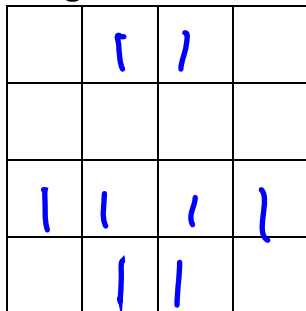
Sad



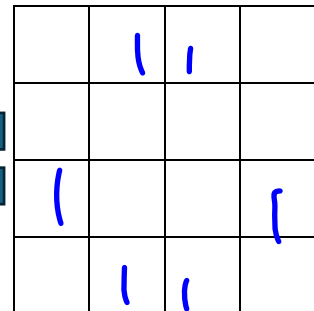
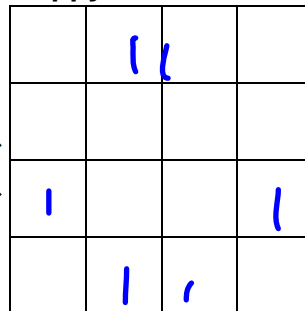
$$\text{Delighted} \times \text{Sad} = \boxed{4}$$

Comparing the same face using maths

Delighted



Happy



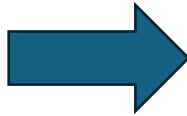
$$\text{Delighted} \times \text{Sad} = \boxed{6}$$

Happy	6		
Sad	2	6	
Delighted	6	4	6
Happy			
Sad			
Delighted			

Going from 2D to 1D

HAPPY

0	1	1	0
0	0	0	0
1	0	0	1
0	1	1	0

[illegible]